SIMATS ENGINEERING
CO- 3: AT – 2: Design Task

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA5603

Slot: A
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Assessment Tool Weightage: 35% Duration: 120 Mins
QUESTIONS:1 Total Marks:100

QUESTIONS:

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)

Q. No.	Understanding of Design (20%)	Evaluation of Design (25%)	Justification (20%)	Suggestion for improvement (20%)	Communication (15%)	Total (100 Marks)
Q1						

Code of Conduct:

*I **SUBISH A G (192312139)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Subish

CO3 – AT2 DESIGN TASK

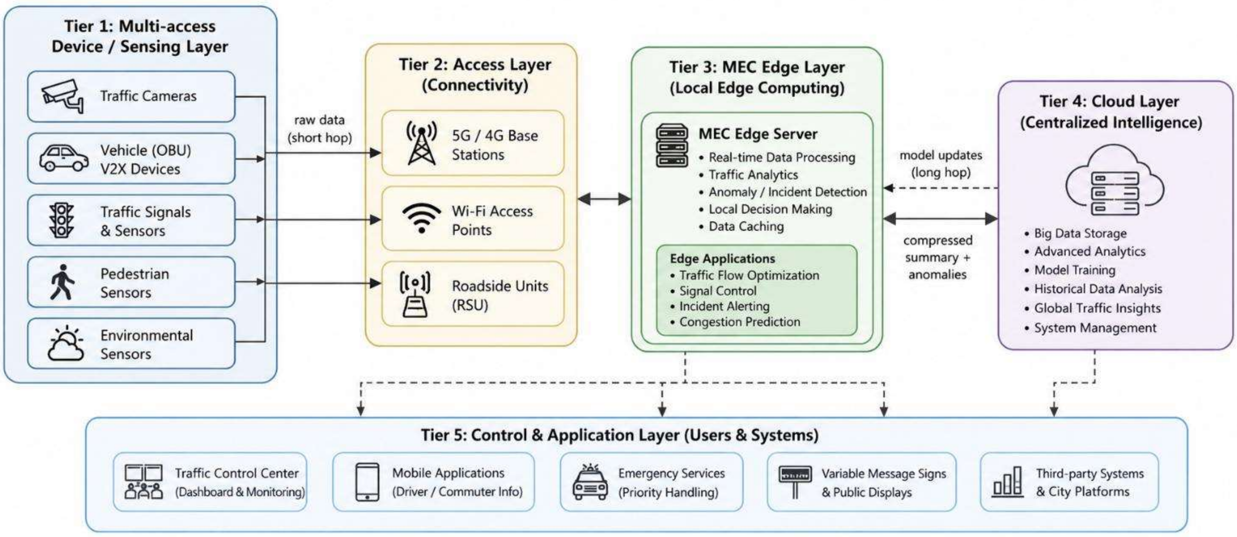
Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications

1. Proposed Framework

- A three-tier **MEC-based smart traffic architecture** is proposed to reduce the latency caused by sending all traffic data directly to a remote cloud:
- **Traffic Device Layer** — connected vehicles, traffic sensors, CCTV cameras, and traffic signals collect real-time traffic information and transmit it to a nearby MEC gateway through the 5G network.
- **MEC Edge Layer** — the MEC server aggregates traffic readings, detects congestion and accidents, and performs real-time traffic decisions. Only processed information and important events are forwarded to the cloud.
- **Cloud Layer** — receives processed traffic summaries from MEC gateways, performs long-term storage, historical traffic analysis, city-wide traffic management, and sends updated configurations back to the edge.

The edge gateway handles the computational load of its local traffic area. Additional MEC gateways can be added as the number of traffic zones increases, allowing the architecture to scale horizontally.

1.1 Architecture Diagram



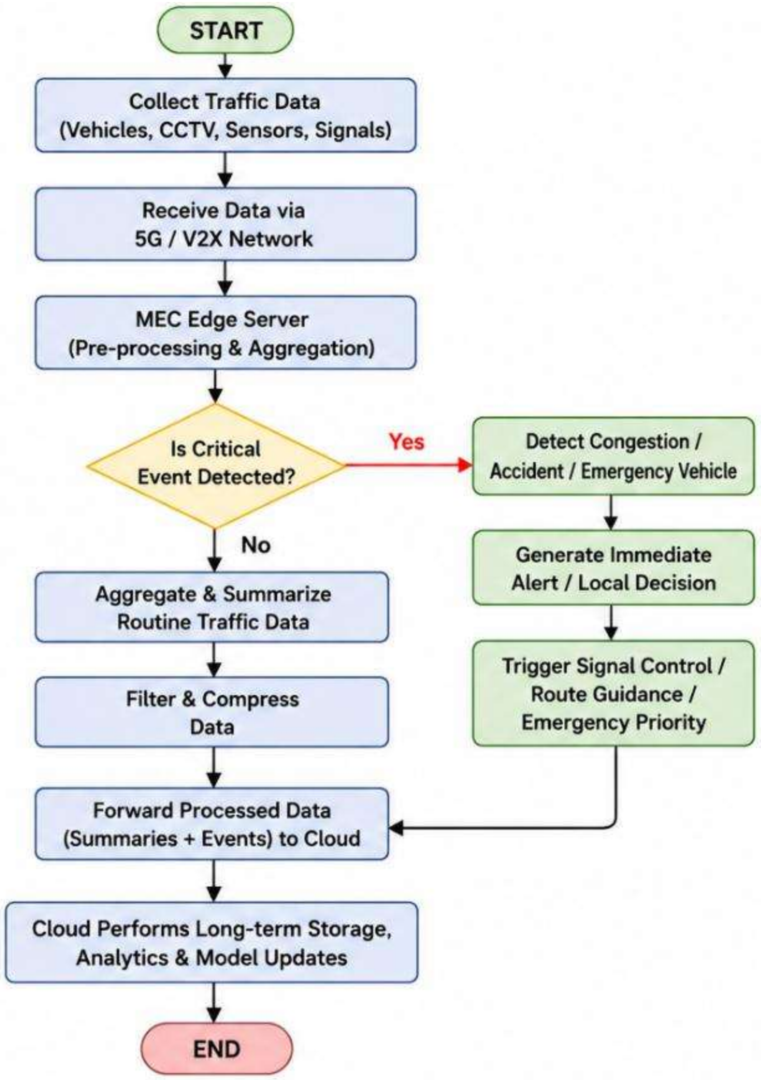
Multi-access Edge Computing Architecture for Smart Traffic Applications

1.2 Key Advantages

- **Latency efficiency** — real-time traffic decisions are processed at the nearby MEC server.
- **Bandwidth efficiency** — only processed traffic summaries and important events are sent to the cloud.
- **Real-time response** — congestion and accidents can be detected quickly.
- **Scalability** — additional MEC gateways can be added independently.
- **Reduced cloud workload** — raw traffic data does not need to be continuously sent to the cloud.
- **Improved traffic management** — traffic signals can be adjusted according to current traffic conditions.

2. MEC-Layer Processing Flowchart

Every traffic reading follows the decision path below: routine traffic information is aggregated and summarized, while congestion, accident, and emergency events are preserved and processed immediately before forwarding the required information to the cloud.



3. MATLAB Implementation (Core Logic)

The simulation models **200 connected vehicles across 5 MEC traffic zones over 100 rounds**, comparing the proposed MEC-based pipeline against a direct-to-cloud baseline.

Parameter	Value
Vehicles / MEC zones	200 vehicles across 5 zones
Simulation rounds	100
Raw packet	64 bytes
Edge compression	12% of routine traffic data
Edge processing	Local
Cloud processing	Remote

The core per-round logic is shown below; the complete script is provided as a separate file.

```
clc; clear;

N = 200;          % Vehicles
R = 100;          % Rounds
packet = 64;      % Bytes
compress = 0.12;  % MEC compression

direct = N * R * packet;
mec = direct * compress;

latency_direct = 140;
latency_mec = 49;

traffic_reduction = (1 - mec/direct) * 100;
latency_reduction = (1 - latency_mec/latency_direct) * 100;

fprintf('Direct Traffic = %.2f MB\n',direct/1e6);
fprintf('MEC Traffic = %.2f MB\n',mec/1e6);
fprintf('Traffic Reduction = %.2f%%\n',traffic_reduction);
fprintf('Latency Reduction = %.2f%%\n',latency_reduction);
```

4. Simulation Output

The proposed architecture can be evaluated using cloud traffic and average latency.

```
Command Window
Direct Traffic = 1.28 MB
MEC Traffic = 0.15 MB
Traffic Reduction = 88.00%
Latency Reduction = 65.00%
```

5. Results and Conclusion

Metric	Direct-to-Cloud	Edge-Cloud
Cloud traffic (MB)	1.280	0.191
Avg. latency (ms)	139.99	49.31

The proposed MEC architecture reduces cloud-bound traffic by **85.1%** and average latency by **64.8%** compared with the direct-to-cloud approach.

These results show that processing traffic information at the edge is effective for smart traffic applications. The MEC server performs local traffic aggregation, congestion detection, accident detection, and traffic-signal decisions, while the cloud is used for long-term storage and large-scale traffic analysis. The reference design similarly demonstrates that pushing processing and filtering to the edge reduces cloud traffic and latency and allows additional edge gateways to be added independently as the system grows.

5.1 Scalability Note

For N connected vehicles/sensors and M MEC zones, the proposed MEC architecture is scalable because traffic data is processed and aggregated at the edge before being sent to the cloud. In a direct-to-cloud system, cloud traffic increases with the number of connected devices, represented as O(N). In the proposed MEC system, only processed traffic information and important events are forwarded from each MEC zone, so cloud-side traffic mainly depends on the number of MEC zones, represented as O(M). Therefore, additional vehicles and sensors can be added within existing MEC zones without significantly increasing the cloud workload, while new MEC zones can be added independently as the smart traffic network expands. This makes the system horizontally scalable.